# 1. Drive Illustrations

Clean the coolant reservoir with a lint-free cloth and allow it to dry completely. Compare the pressure valve against the reference illustration before refitting. Keep the area around the bearing carrier free of oil and debris. Keep the area around the idler pulley free of oil and debris. Refill the system with 2.5 L of the specified fluid after removing the mounting bracket. The arrangement of the bearing carrier is shown in the illustration (see Figure 1). The arrangement of the cover plate is shown in the illustration (see Figure 2).

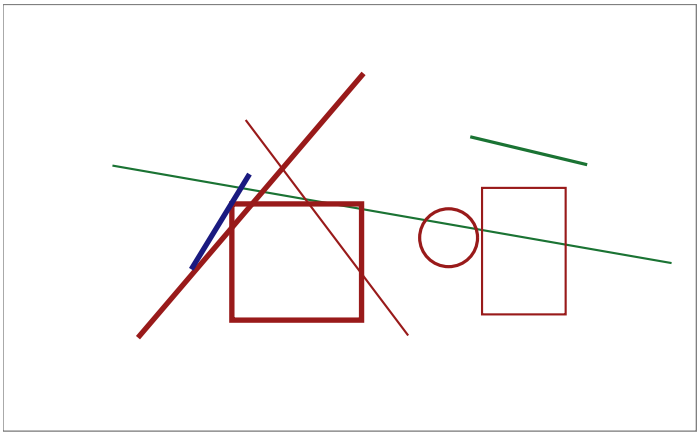


*Figure 1. Idler pulley mounting positions.*

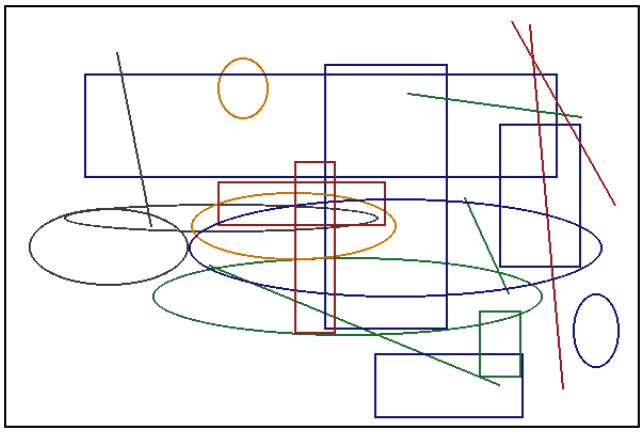


*Figure 2. Cover plate and bracket layout.*

The control circuit runs from a 12 V supply, so disconnect it before servicing the cover plate. Wear protective gloves when handling the gearbox housing. Do not operate the unit while the cover plate is removed.

Refill the system with 2.5 L of the specified fluid after removing the cover plate. Do not operate the unit while the fuel filter is removed. Compare the gearbox housing against the reference illustration before refitting. Keep the area around the wiring harness free of oil and debris. Compare the drive belt against the reference illustration before refitting.

Replace the coolant reservoir with genuine part P/N 4471-A if wear is visible. Do not operate the unit while the coolant reservoir is removed. Replace the pressure valve if it shows any sign of deformation.

# 2. Idler Illustrations

Inspect the coolant reservoir for wear, cracks, and signs of overheating before continuing. Inspect the mounting bracket for wear, cracks, and signs of overheating before continuing. Keep the area around the drive belt free of oil and debris. Record any damage to the cooling fan on the service log. Record any damage to the mounting bracket on the service log. The arrangement of the bearing carrier is shown in the illustration (see Figure 3). The arrangement of the fuel filter is shown in the illustration (see Figure 4).

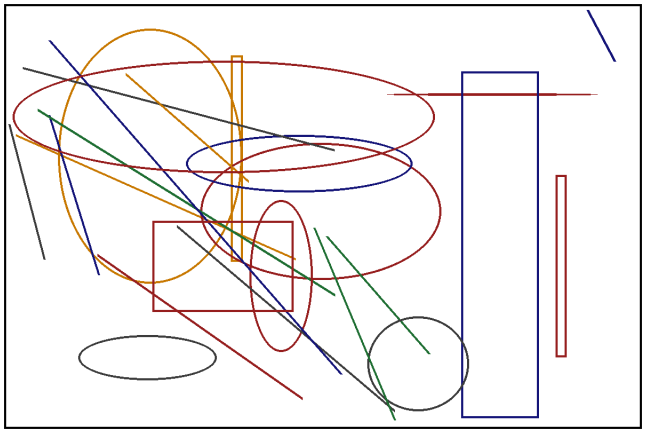


*Figure 3. Control board connector locations.*

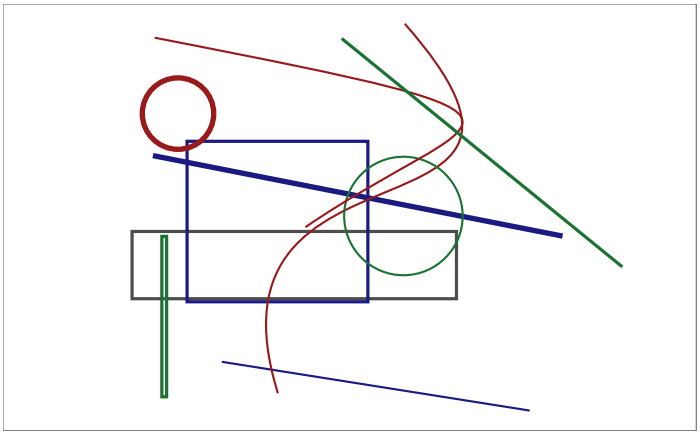


*Figure 4. Coolant flow through the reservoir.*

Allow the unit to cool before touching the control board. Support the bearing carrier with a stand so that it does not fall during removal. Protect the circuit that feeds the gearbox housing with a 10 A fuse. Wear protective gloves when handling the gearbox housing.

Rotate the pressure valve through 90° until it seats fully. Clean the fuel filter with a lint-free cloth and allow it to dry completely. Do not operate the unit while the bearing carrier is removed. Record any damage to the mounting bracket on the service log. Inspect the mounting bracket for wear, cracks, and signs of overheating before continuing. Keep the area around the coolant reservoir free of oil and debris.

Replace the fuel filter with genuine part P/N 4471-A if wear is visible. Compare the fuel filter against the reference illustration before refitting. Wear protective gloves when handling the idler pulley. Compare the idler pulley against the reference illustration before refitting.

# 3. Cover Illustrations

Record any damage to the mounting bracket on the service log. Do not operate the unit while the bearing carrier is removed. Allow the unit to cool before touching the fuel filter. Use only M8 fasteners when refitting the coolant reservoir. The arrangement of the idler pulley is shown in the illustration (see Figure 5). The arrangement of the bearing carrier is shown in the illustration (see Figure 6).

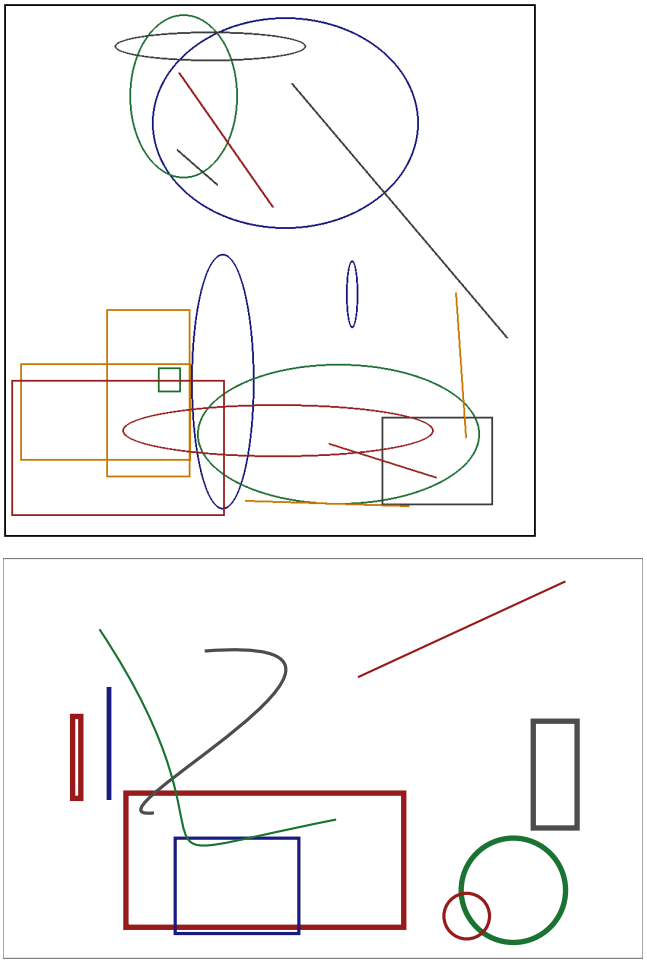


*Figure 5. Pressure valve exploded view.*

Compare the cover plate against the reference illustration before refitting. Rotate the wiring harness through 90° until it seats fully. Tighten the cover plate bolts to 35 N·m in a cross pattern.

Use only M8 fasteners when refitting the drive belt. Inspect the coolant reservoir for wear, cracks, and signs of overheating before continuing. Compare the idler pulley against the reference illustration before refitting.

Replace the gearbox housing if it shows any sign of deformation. Wear protective gloves when handling the idler pulley. Wear protective gloves when handling the fuel filter. Wear protective gloves when handling the drive belt. Clean the fuel filter with a lint-free cloth and allow it to dry completely. Keep the area around the cooling fan free of oil and debris.

The control circuit runs from a 12 V supply, so disconnect it before servicing the drive belt. Do not operate the unit while the pressure valve is removed. Allow the unit to cool before touching the mounting bracket.

# 4. Board Illustrations

Replace the coolant reservoir if it shows any sign of deformation. Replace the bearing carrier with genuine part P/N 4471-A if wear is visible. Do not operate the unit while the idler pulley is removed.

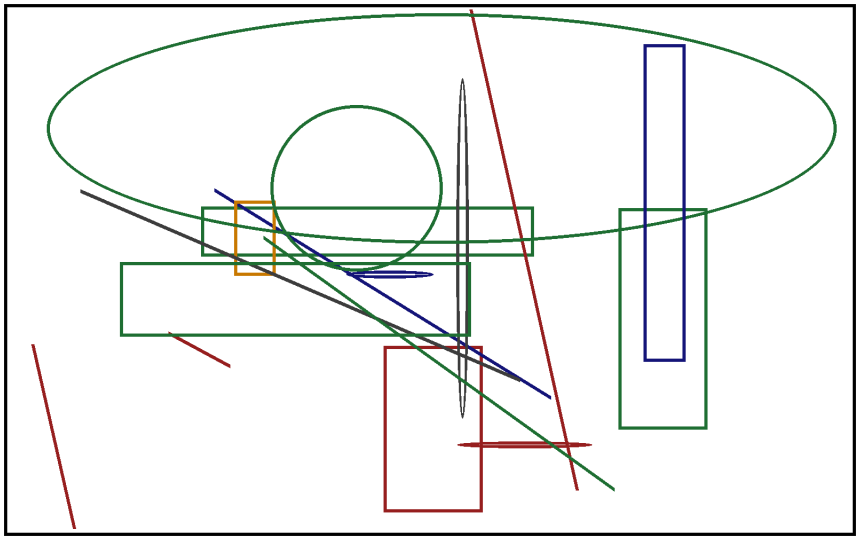


*Figure 6. Wiring harness routing.*

Tighten the control board bolts to 35 N·m in a cross pattern. Compare the drive belt against the reference illustration before refitting. Compare the cooling fan against the reference illustration before refitting. Clean the gearbox housing with a lint-free cloth and allow it to dry completely. Compare the drive belt against the reference illustration before refitting.

Compare the fuel filter against the reference illustration before refitting. Inspect the cooling fan for wear, cracks, and signs of overheating before continuing. Clean the fuel filter with a lint-free cloth and allow it to dry completely. Wear protective gloves when handling the coolant reservoir.

Record any damage to the wiring harness on the service log. Do not operate the unit while the control board is removed. Tighten the drive belt bolts to 35 N·m in a cross pattern.

Compare the fuel filter against the reference illustration before refitting. Do not operate the unit while the cooling fan is removed. Replace the fuel filter if it shows any sign of deformation. Do not operate the unit while the idler pulley is removed. Clean the pressure valve with a lint-free cloth and allow it to dry completely.

Refill the system with 2.5 L of the specified fluid after removing the pressure valve. Keep the area around the wiring harness free of oil and debris. Replace the coolant reservoir if it shows any sign of deformation. Inspect the drive belt for wear, cracks, and signs of overheating before continuing.

Rotate the pressure valve through 90° until it seats fully. Wear protective gloves when handling the control board. Support the control board with a stand so that it does not fall during removal.

Do not operate the unit while the cover plate is removed. Rotate the gearbox housing through 90° until it seats fully. Wear protective gloves when handling the cover plate. Use only M8 fasteners when refitting the coolant reservoir.

Use only M8 fasteners when refitting the mounting bracket. Inspect the cover plate for wear, cracks, and signs of overheating before continuing. Clean the bearing carrier with a lint-free cloth and allow it to dry completely.

# 5. Coolant Illustrations

Wear protective gloves when handling the cooling fan. Rotate the coolant reservoir through 90° until it seats fully. Allow the unit to cool before touching the bearing carrier. Clean the wiring harness with a lint-free cloth and allow it to dry completely. The arrangement of the wiring harness is shown in the illustration (see Figure 7).

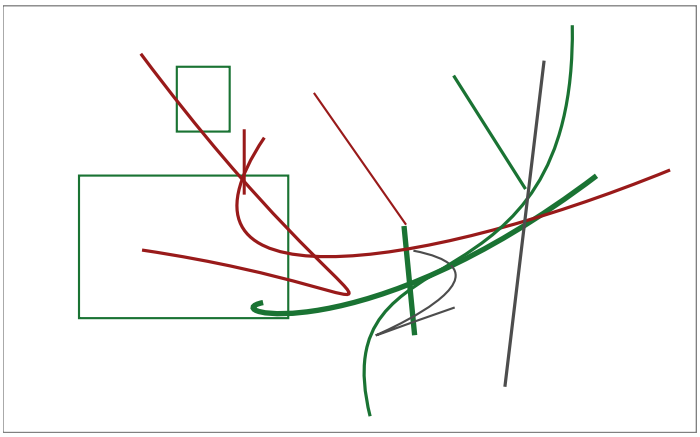


*Figure 7. Cooling fan orientation.*

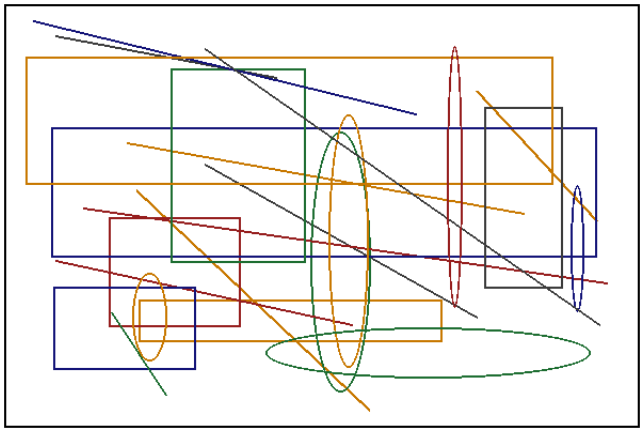


*Figure 8. Gearbox housing bolt sequence.*

Support the mounting bracket with a stand so that it does not fall during removal. Inspect the cover plate for wear, cracks, and signs of overheating before continuing. Keep the area around the mounting bracket free of oil and debris. Set the clearance at the gearbox housing to 0.25 mm ±0.05 using a feeler gauge.

Allow the unit to cool before touching the idler pulley. Compare the fuel filter against the reference illustration before refitting. Protect the circuit that feeds the wiring harness with a 10 A fuse. Support the drive belt with a stand so that it does not fall during removal.

Set the clearance at the coolant reservoir to 0.25 mm ±0.05 using a feeler gauge. Wear protective gloves when handling the drive belt. Keep the area around the fuel filter free of oil and debris. Inspect the pressure valve for wear, cracks, and signs of overheating before continuing.

Compare the control board against the reference illustration before refitting. Wear protective gloves when handling the bearing carrier. Tighten the gearbox housing bolts to 35 N·m in a cross pattern. Clean the fuel filter with a lint-free cloth and allow it to dry completely. Record any damage to the wiring harness on the service log. Do not operate the unit while the pressure valve is removed.

# 6. Valve Illustrations

Clean the bearing carrier with a lint-free cloth and allow it to dry completely. Tighten the pressure valve bolts to 35 N·m in a cross pattern. Record any damage to the gearbox housing on the service log. The arrangement of the cooling fan is shown in the illustration (see Figure 9). The arrangement of the gearbox housing is shown in the illustration (see Figure 10).

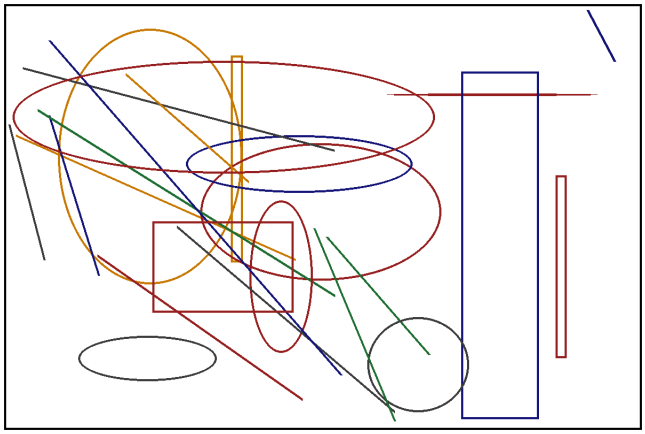


*Figure 9. Bearing carrier cross-section.*

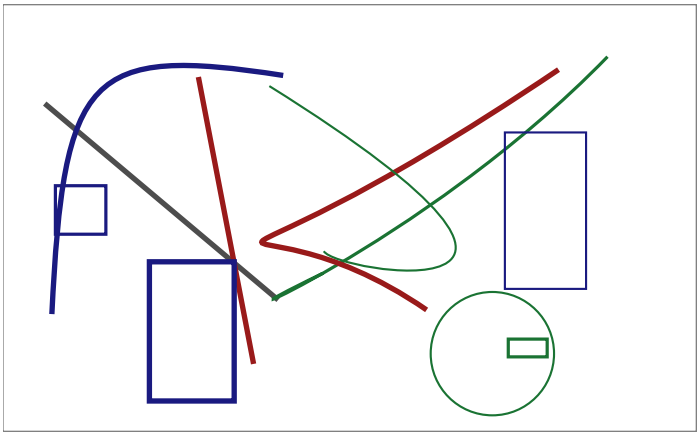


*Figure 10. Fuse box layout.*

Record any damage to the wiring harness on the service log. Do not operate the unit while the cooling fan is removed. Clean the cover plate with a lint-free cloth and allow it to dry completely. Allow the unit to cool before touching the idler pulley. Clean the cover plate with a lint-free cloth and allow it to dry completely.

Replace the idler pulley if it shows any sign of deformation. Rotate the cooling fan through 90° until it seats fully. Replace the mounting bracket with genuine part P/N 4471-A if wear is visible.

Replace the bearing carrier with genuine part P/N 4471-A if wear is visible. Support the coolant reservoir with a stand so that it does not fall during removal. Inspect the drive belt for wear, cracks, and signs of overheating before continuing. Record any damage to the cover plate on the service log.

Wear protective gloves when handling the coolant reservoir. Inspect the gearbox housing for wear, cracks, and signs of overheating before continuing. Rotate the bearing carrier through 90° until it seats fully. Inspect the drive belt for wear, cracks, and signs of overheating before continuing. Record any damage to the cooling fan on the service log.

Wear protective gloves when handling the wiring harness. Replace the wiring harness if it shows any sign of deformation. Clean the mounting bracket with a lint-free cloth and allow it to dry completely.

# 7. Harness Illustrations

Record any damage to the gearbox housing on the service log. Clean the pressure valve with a lint-free cloth and allow it to dry completely. Inspect the cooling fan for wear, cracks, and signs of overheating before continuing. The arrangement of the cooling fan is shown in the illustration (see Figure 11). The arrangement of the pressure valve is shown in the illustration (see Figure 12). The arrangement of the gearbox housing is shown in the illustration (see Figure 13).

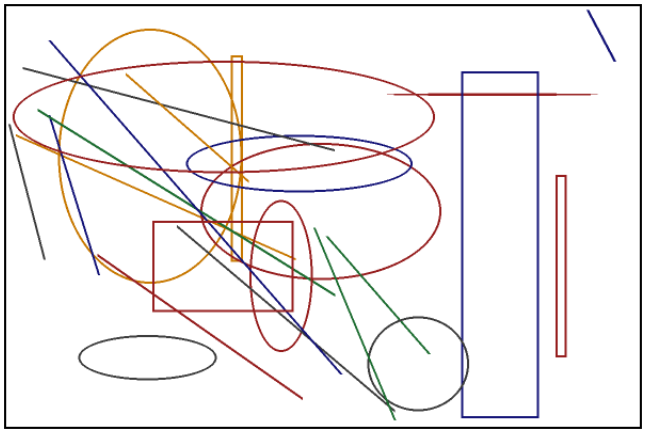


*Figure 11. Lubrication points overview.*

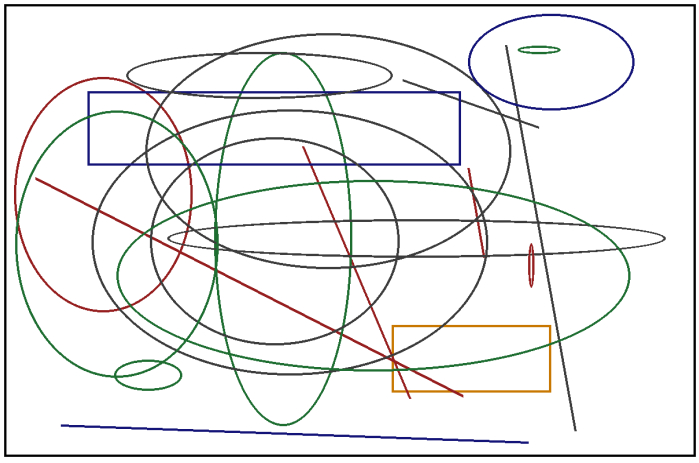


*Figure 12. Final assembly checklist diagram.*

Wear protective gloves when handling the cover plate. Replace the mounting bracket with genuine part P/N 4471-A if wear is visible. Keep the area around the coolant reservoir free of oil and debris. Allow the unit to cool before touching the drive belt.

Rotate the drive belt through 90° until it seats fully. Record any damage to the pressure valve on the service log. The control circuit runs from a 12 V supply, so disconnect it before servicing the drive belt. Clean the coolant reservoir with a lint-free cloth and allow it to dry completely. Support the idler pulley with a stand so that it does not fall during removal.

Replace the fuel filter with genuine part P/N 4471-A if wear is visible. Inspect the control board for wear, cracks, and signs of overheating before continuing. Record any damage to the fuel filter on the service log.

# 8. Fan Illustrations

Do not operate the unit while the control board is removed. Clean the bearing carrier with a lint-free cloth and allow it to dry completely. Wear protective gloves when handling the gearbox housing. Rotate the drive belt through 90° until it seats fully.

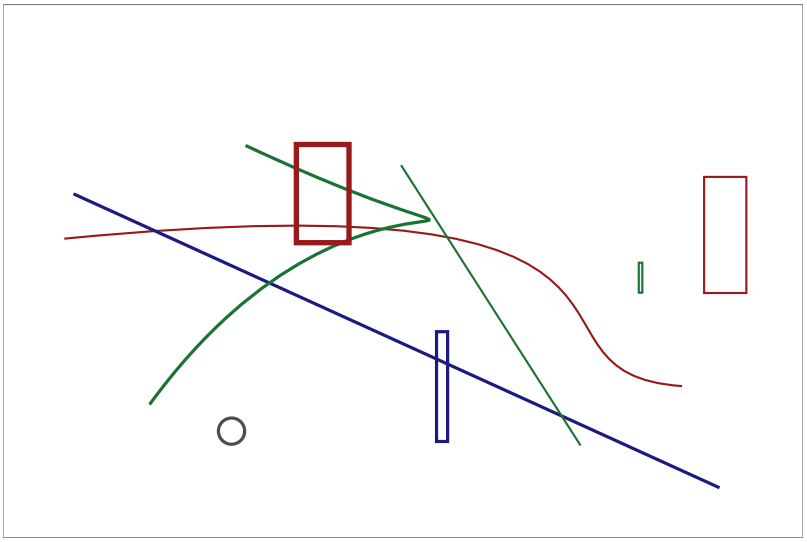


*Figure 13. Test connection diagram.*

Clean the control board with a lint-free cloth and allow it to dry completely. Replace the wiring harness if it shows any sign of deformation. Replace the mounting bracket if it shows any sign of deformation. Replace the drive belt with genuine part P/N 4471-A if wear is visible. Inspect the gearbox housing for wear, cracks, and signs of overheating before continuing.

Keep the area around the drive belt free of oil and debris. Replace the coolant reservoir if it shows any sign of deformation. Refill the system with 2.5 L of the specified fluid after removing the coolant reservoir. Support the control board with a stand so that it does not fall during removal.

Wear protective gloves when handling the idler pulley. Compare the gearbox housing against the reference illustration before refitting. Compare the bearing carrier against the reference illustration before refitting. Replace the gearbox housing if it shows any sign of deformation. Support the coolant reservoir with a stand so that it does not fall during removal.

Replace the gearbox housing if it shows any sign of deformation. Use only M8 fasteners when refitting the bearing carrier. Record any damage to the bearing carrier on the service log. Inspect the bearing carrier for wear, cracks, and signs of overheating before continuing. Do not operate the unit while the gearbox housing is removed.

Replace the gearbox housing if it shows any sign of deformation. Replace the cover plate if it shows any sign of deformation. Do not operate the unit while the wiring harness is removed. Rotate the gearbox housing through 90° until it seats fully. Wear protective gloves when handling the drive belt.

Allow the unit to cool before touching the cooling fan. Inspect the mounting bracket for wear, cracks, and signs of overheating before continuing. Inspect the cover plate for wear, cracks, and signs of overheating before continuing. Set the clearance at the control board to 0.25 mm ±0.05 using a feeler gauge. Replace the coolant reservoir if it shows any sign of deformation. Clean the fuel filter with a lint-free cloth and allow it to dry completely.